umbra.

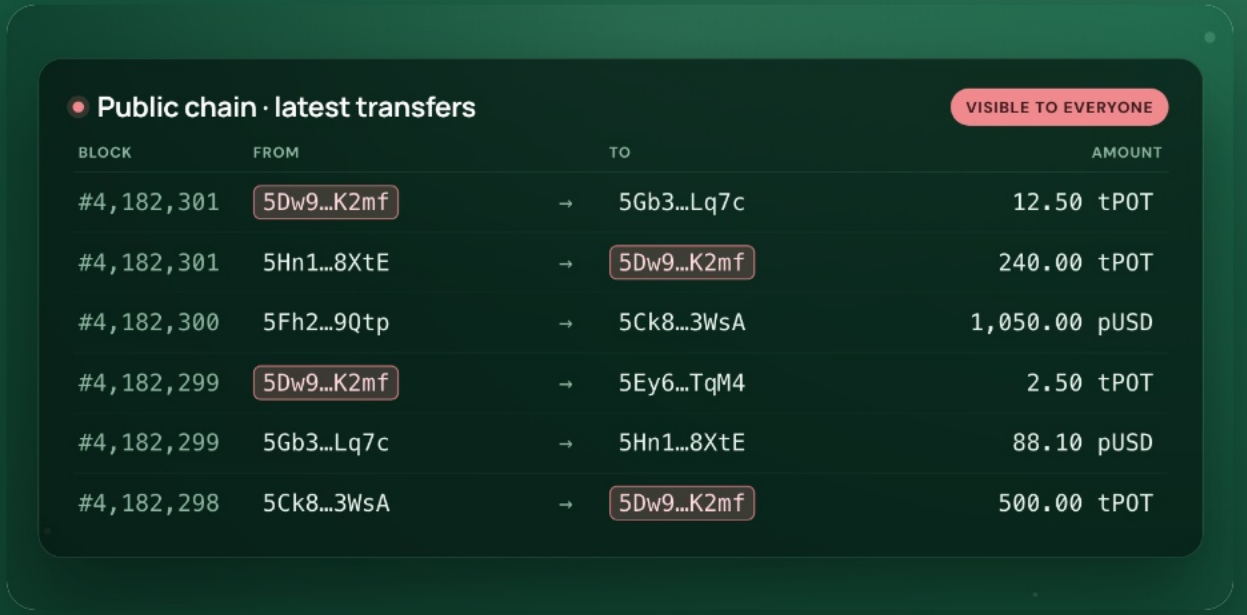
Pay anyone.
Hide everything.

A zero-knowledge privacy layer for payments, payment requests and swaps on Portaldot V3.

Live on Portaldot V3 testnet | tPOT + pUSD | Portaldot Hacker House 2026

Every payment on a public chain is public forever.

- One reused wallet is an open diary: a single payment exposes your salary, rent, savings and every shop you use.
- Businesses cannot pay staff, suppliers or customers without publishing their margins and treasury.
- So real commerce stays off-chain, and a payments chain stalls at the first serious use case.



One wallet, traced across every transfer

Anyone can follow it, forever.

A private balance for every Portaldot user.

1

Deposit once

From the Portaldot Extension or any EVM wallet. Public, and the only step your wallet signs.

2

Use it privately

Pay, request, split bills and swap tPOT ↔ pUSD. Proofs are made in your browser; a relayer submits them.

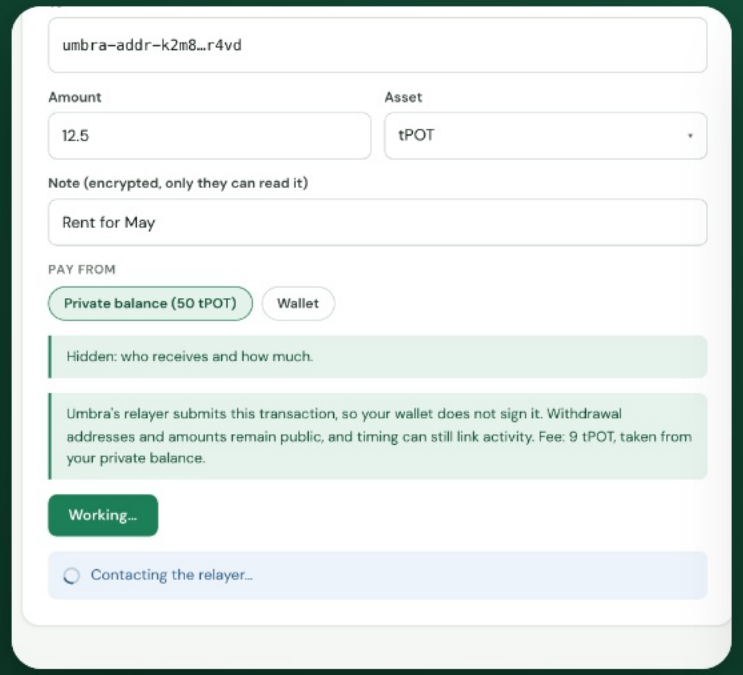
3

Withdraw anytime

To any address. A zero-knowledge proof shows you own funds in the pool without revealing which.

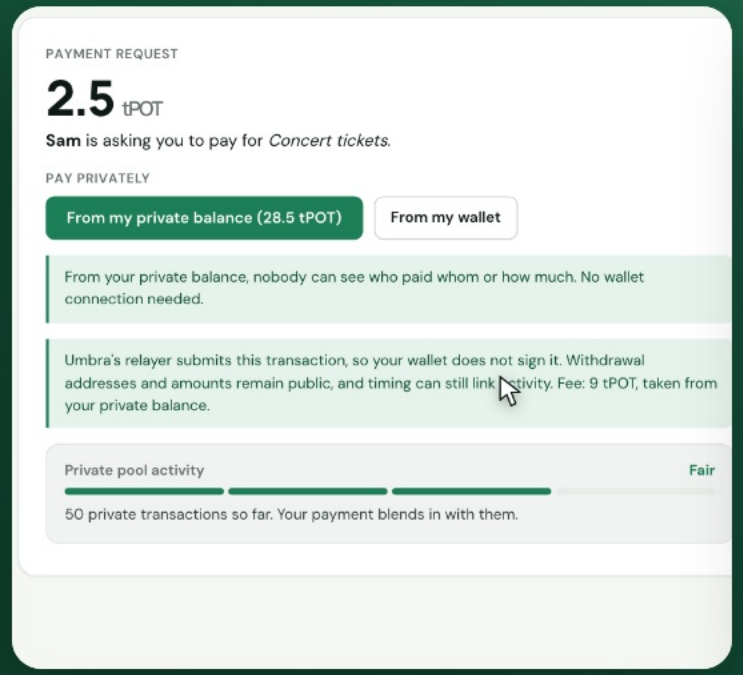
- One shared pool for tPOT and pUSD
- Encrypted notes and memos
- Payment links and QR codes
- Read-only viewing keys for auditors

A real app, running today.



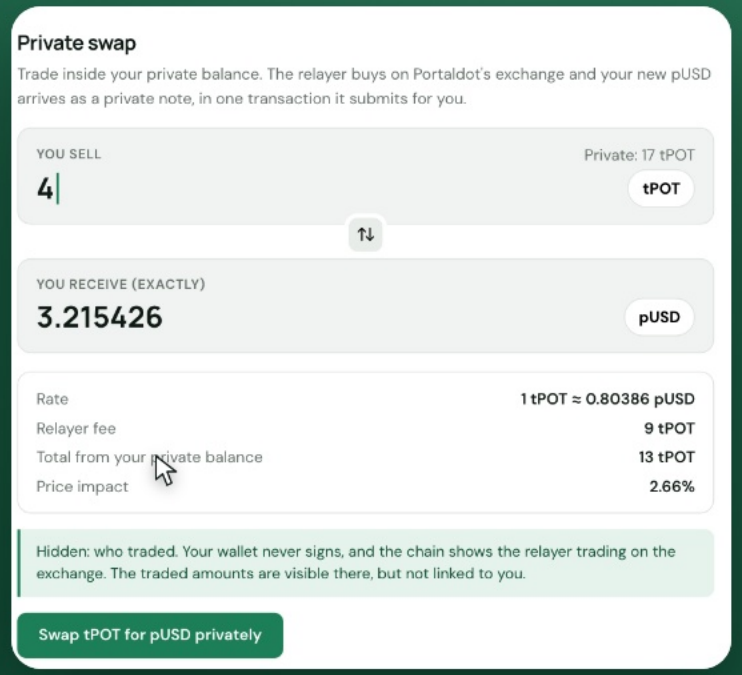
Send privately

Encrypted note; a relayer submits it, so your wallet never signs.



Get paid with a link

Open a ?pay= link, see who and what for, pay privately.

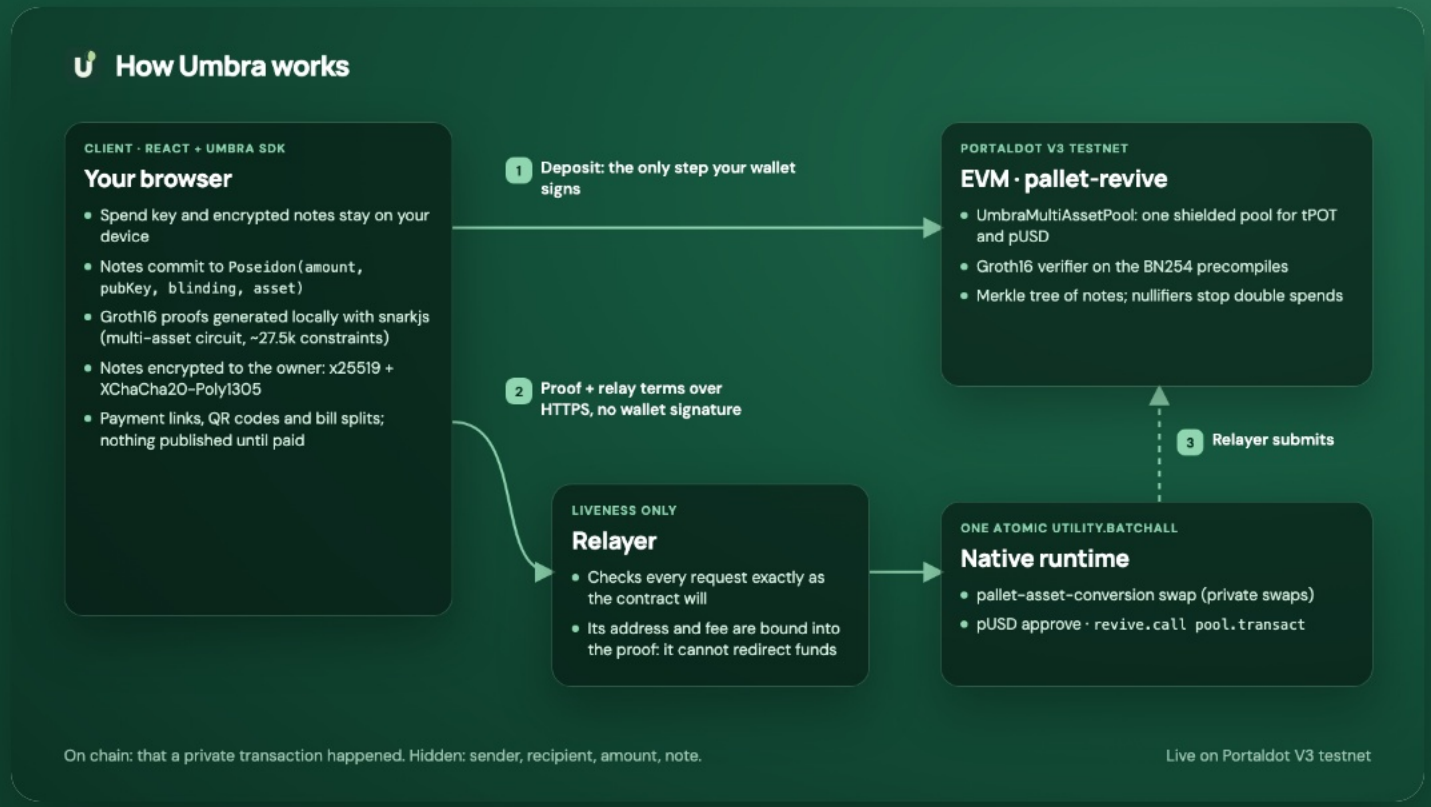


Swap tPOT ↔ pUSD

Live quote, fee and price impact. It fills exactly or not at all.

Proofs in the browser. Verified on chain.

- Balance = encrypted notes; each is a Poseidon commitment in an on-chain Merkle tree.
- Groth16 join-split proofs (~27.5k constraints) check ownership and value conservation.
- The relayer is bound into the proof: it can delay, never redirect funds.



Built on what Portaldot V3 provides natively.

EVM + WASM + native pallets

One chain, three runtimes. Pools and verifiers run in EVM; a second verifier runs in ink!; swaps use native pallets.

BN254 precompiles

On-chain Groth16 verification through `0x06–0x08`: about 0.035 s of block weight per proof.

Native accounts call EVM

`revive.call` lets Portaldot Extension users deposit with their 5… account. No EVM wallet needed.

pUSD as a first-class asset

The pallet-assets ERC-20 precompile shields pUSD (#7777): confidential stablecoin payments.

Native DEX, atomic batches

Private swaps settle on `pallet-asset-conversion` inside one `utility.batchAll`: it fills or nothing happens.

Designed for its economics

Redesigned for Portaldot's storage deposits: fixed-pool deposit 8.96 → 0 tPOT, private transaction 12.8 → 7.68 tPOT.

- Portaldot's whitepaper lists a **ZKP-based privacy protection suite** for compliant private commerce. Umbra ships it on the live chain.

Live on testnet. Verified end to end.

75
Foundry contract tests, real Poseidon

7
WASM verifier tests on the same real proofs

30
end-to-end checks with real Groth16 proofs

23
relayer checks: tampering, replays, fees

BLOCK **169,320**

Deployment begins
Poseidon hasher, deployed from the Portaldot Extension (revive.instantiateWithCode)

`ext 0x81c2d632…e9f7f`

BLOCK **215,672**

Deposit into the private pool
Signed by the user's own account, revive.call

`ext 0xb6ebef2a…bacd`

BLOCK **215,680**

First private swap, relayed
One atomic utility.batchAll. The trader's wallet never signed

`ext 0x0f6c108f…9af4`

Plus live read-only suites against the deployed pool, including a real proof accepted by the deployed verifier. Open source: **github.com/drained69/umbra**

Clear about what is hidden, and what is not.

ACTION	HIDDEN	VISIBLE ON CHAIN
Deposit	Which notes it becomes	Depositor, asset, amount
Private payment	Sender, recipient, amount, note	That a private tx happened; the relayer
Private swap	Who traded	The relayer's trade on the exchange
Withdrawal	Which deposit it came from	Amount and receiving address
Request link	Everything, until paid	Nothing

KNOWN LIMITS AND THE PLAN

- Not independently audited. Two internal reviews, fixes covered by regression tests. Test tokens only.
- Single-contributor trusted setup, for demo use. Production needs a multi-party ceremony.
- Privacy grows with the crowd; a live meter in the app shows pool cover.
- Contracts are immutable with no admin keys. The relayer can censor, never steal.

More than an app: a privacy primitive for the ecosystem.

Reusable by every dApp

A multi-asset shielded pool any Portaldot asset can join, plus a TypeScript SDK and a relayer other apps can plug into.

Private, yet compliant

Read-only viewing keys give accountants and auditors visibility without spend power. That is what real businesses need.

Built-in revenue

Relayer fees on private transactions and a spread on private swaps pay for the service. Every new user and asset deepens everyone's privacy.

Why it compounds: one anonymity set across tPOT, pUSD and future assets means each added user makes every payment harder to trace.

From a working testnet beta to real-funds privacy.

TODAY

- Live on V3 testnet with real proofs
- Private payments, requests, bill splits, swaps
- Viewing keys, relayer, SDK, tests

NEXT

- Batched private swaps that hide amounts
- Multi-party trusted-setup ceremony
- Independent professional audit
- Deploy the WASM second verifier

LATER

- Cross-chain transfers, Ethereum Sepolia ↔ Portaldot (prototype built)
- V3 Live: same EVM-capable runtime
- More Portaldot assets in the shared pool

Support turns the demo into infrastructure: an audit, a real ceremony and relayer liquidity are the steps between this testnet beta and production.

Pay anyone.
Hide everything.

LIVE APP	umbrav1.up.railway.app
DEMO VIDEO	youtube.com/watch?v=hMTj8YFXGWM
SOURCE CODE	github.com/drained69/umbra

Portaldot V3 testnet · test tokens only · MIT licensed

Scan to try it live

Portaldot Extension + testnet tPOT